Step 5C: Elastic Net Robustness Check

Elastic Net is not a latent-variable method. It is included here as a robustness check on the interaction-augmented candidate set: if a variable or interaction is important in PLS/VIP and also survives penalized regression, that is stronger evidence that it belongs in the final story.

Best Elastic Net Setting

- Alpha: 0.03
- L1 ratio: 0.5
- Mean selected variables across folds: 27.000
- CV R2 on scaled turnout: 0.617
- CV RMSE on scaled turnout: 0.619

Selected Variables

Variable	Family	Abs scaled coef	Direction
block3_visible_minority_share	immigration_citizenship	0.291	lower turnout
block1_bachelors_or_higher_25_64_share	household_class	0.202	higher turnout
block1_age_18_34_share	age_structure	0.180	lower turnout
block2_apartment_share	housing_form_density	0.125	higher turnout
block1_bachelors_or_higher_25_64_share__x__block4_effective_mayoral_candidates_5pct	other	0.123	higher turnout
block1_average_household_size	household_class	0.114	lower turnout
block2_population_density_per_km2	housing_form_density	0.104	higher turnout
block1_low_income_lim_at_share__x__block5_requests_311_per_1000	other	0.088	higher turnout
block3_non_citizen_share	immigration_citizenship	0.086	lower turnout
block4_federal_margin	federal_competitiveness	0.077	higher turnout
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	0.069	lower turnout
block3_non_official_mother_tongue_share	immigration_citizenship	0.068	lower turnout
block5_ksi_collision_events_2021_2025_per_1000__x__block1_unemployment_rate_share	other	0.066	lower turnout
block1_low_income_lim_at_share	household_class	0.059	lower turnout
block5_community_centre_access_1200m	service_access	0.052	lower turnout
block1_average_household_size__x__block5_ksi_collision_events_2021_2025_per_1000	other	0.049	lower turnout
block2_condo_share	housing_form_density	0.043	higher turnout
block1_age_65_plus_share__x__block1_average_household_size	other	0.040	lower turnout
block1_low_income_lim_at_share__x__block1_age_35_64_share	other	0.036	lower turnout
block1_average_household_size__x__block1_age_35_64_share	other	0.032	higher turnout

Variable	Family	Abs scaled coef	Direction
block2_renter_share	housing_tenure	0.032	lower turnout
block5_shelter_access_1200m	service_access	0.026	lower turnout
block5_library_access_1200m	service_access	0.010	lower turnout
block5_requests_311_per_1000	service_contact	0.008	higher turnout
block5_social_housing_share	service_contact	0.004	higher turnout

Agreement With Interaction-Augmented PLS

Top interaction-augmented PLS variables also selected by Elastic Net:

- block1_average_household_size
- block1_bachelors_or_higher_25_64_share
- block1_bachelors_or_higher_25_64_share__x__block4_effective_mayoral_candidates_5pct
- block3_non_citizen_share
- block3_non_official_mother_tongue_share
- block3_visible_minority_share
- block4_effective_mayoral_candidates_5pct
- block4_federal_margin
- block4_mayoral_top_two_margin

Interpretation

Elastic Net should not replace PLS for this project because it does not create latent components. Its value is diagnostic. Variables and interactions that survive both methods are sturdy candidates for the final narrative. Terms that are high-VIP but not selected by Elastic Net may still matter as part of a latent geography, but they are less convincing as standalone predictors.

The strongest Elastic Net terms in plain language are visible minority share, bachelor or higher education share, age 18 to 34 share, apartment share, bachelor or higher education share with effective mayoral candidates above five percent, average household size, population density, low income share with 311 requests per 1000 residents, non citizen share, federal margin, effective mayoral candidates above five percent, non official mother tongue share. As a robustness check, this supports reference categories around racialized geography, education, young adults, apartment and density context, local electoral fragmentation, household size, low income and 311 service contact, citizenship, and federal competitiveness.